Khushi Singhal

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Indian Institute of Technology Roorkee

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Education

Degree/Board	Institution	Year	CGPA/%
B.Tech., Electrical Engineering	Indian Institute of Technology Roorkee	2026	8.562/10
Class XII (CBSE)	Jayshree Periwal High School, Jaipur	2022	95.20%
Class X (CBSE)	Maheshwari Girls Public School, Jaipur	2020	97.80%

Scholastic and Programming Achievements

- All India Rank 3,406, JEE Advanced 2022.
- All India Rank 7,425, JEE Main 2022.

Work Experience

Google July 2026–Present

Software Engineer

- Built a Go heartbeat reporter publishing job status, progress, and gRPC error codes from worker VMs to GCS.
- Computed monotonic, byte-based job progress and restored job state from GCS after VM restarts.
- Implemented a Parquet export stage converting SQLite records into a typed schema for data-warehouse ingestion.
- Designed a reusable GCS client interface with an in-memory fake for dependency injection and hermetic tests.
- Defined a Protobuf schema for backup metadata and declarative job-config validation rules to fail fast.
- Joined production support rotation and built an AI-agent workflow to triage and merge duplicate alert tickets.

Tech stack: Go, C++, Protocol Buffers, SQLite, Parquet, Google Cloud Storage, Linux

Google May–Aug 2025

Software Engineering Intern

- Implemented transactional data updates in a distributed pipeline to ensure data consistency.
- Used a feature flag to control migration from blind writes to Read-Modify-Write for fresher data.
- Optimized mutation batching and built a reusable, validated prototype for other teams.

Tech stack: Java, Protocol Buffers, internal data-processing and testing libraries

Relevant Coursework

Data Structures & Algorithms, Programming in C++, Microprocessors, Google Cloud Computing Foundations.

Skills

Programming: C/C++, Go, Python, Java, JavaScript, HTML, MATLAB, Verilog, SQL.
Systems/Data: Protocol Buffers, SQLite, Parquet, Google Cloud Storage.
Tools: Linux Shell Utilities, Git, Visual Studio, LTSpice, Simulink, MS Office.

Projects

Power System State Estimator Jan–Apr 2025

Electrical Department, IIT Roorkee

- Implemented Weighted Least Squares in MATLAB for secure parameter analysis and ported it to JavaScript for real-time web processing.
- Built a virtual-lab interface for file uploads, computation, and table/heatmap visualization while preserving data privacy.

Tech stack: MATLAB, JavaScript, HTML

RISC-V Simulator Nov–Dec 2024

Self Project

- Developed a 5-stage pipelined RISC-V processor covering fetch, decode, execute, memory, and write-back stages.
- Designed a cache simulator to improve memory-access efficiency and overall performance.

Tech stack: C++

ReadSleuth Dec 2023–Jan 2024

MDG Space, IIT Roorkee

- Built a web app for text- and image-based PDF question search using Pytesseract and PyMuPDF for extraction.
- Used SBERT embeddings for similarity search and Flask with MongoDB for PDF storage and management.

Tech stack: Python, HTML, CSS, Flask, MongoDB

IngredientRecipe Guide Dec 2023–Jan 2024

Vision and Language Group, IIT Roorkee

- Built an ML system to recognize food ingredients and recommend recipes, with Python-based image extraction and conversion.
- Augmented data using Bing Image Downloader and Keras; applied GroundedSAM and YOLOv8 for annotation and detection.

Tech stack: Python

Positions of Responsibility & Extracurriculars

Undergraduate Teaching Assistant Jul 2025–Present

Programming with C++, IIT Roorkee

- Supported EEC-101 labs, answered coding questions, and evaluated work through code reviews and viva.

Competitive Programming Jul 2023–Present

Online Judges

- Active on Codeforces, CodeChef, InterviewBit, and LeetCode.

Participant, Syntax Error X Oct 2023

Software Development Section, IIT Roorkee

- Developed a Python 2D platformer with collision mechanics, animation states, dynamic obstacles, and varied movement.